

Rajat Malik

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PROFESSIONAL SUMMARY

Final-year B.E. (AI & ML) student at Chandigarh University with a strong foundation in building real-world AI applications — from autonomous email agents and multimodal research engines to collaborative platforms with semantic memory. Experienced in LLM orchestration, privacy-first local inference, and full-stack deployment. Proven ability to ship under constraints with a focus on performance, security, and reproducibility.

EDUCATION

Chandigarh University Jun 2023 – Jun 2027

B.E. — Computer Science (AI & ML) | CGPA: 7.6 / 10.0 Mohali, Punjab, India

Relevant Coursework: Machine Learning, Deep Learning, DSA, Operating Systems, DBMS, Computer Networks, NLP

SKILLS SUMMARY

Languages: Python, TypeScript, JavaScript (ES6+), SQL, C++

LLM & Inference: Ollama, GGUF Quantisation, LangChain, LangGraph, HuggingFace

Databases: Neo4j, ChromaDB, Pinecone, FAISS, Qdrant, MongoDB

AI / ML: Latent Reasoning, JEPA, Agentic Systems, RAG, GraphRAG

Frameworks: React.js, Node.js, FastAPI, Flask, Socket.IO, PyTorch, TensorFlow

Tools: Docker, GCP (Compute Engine, App Engine, Firebase), Git, CI/CD

Soft Skills: Technical Leadership · Cross-functional Collaboration · Problem-solving Under Pressure · Research Communication · Ownership Mindset · Adaptability

EXPERIENCE

EDSHODH LLP Jun 2025 – Dec 2025

Python Developer Intern (Full-time) Remote

- Architected autonomous AI agents for internal process automation using LangChain; owned end-to-end pipeline design from prompt engineering to evaluation and deployment.
- Optimised Python inference scripts for edge deployment via GGUF quantisation and Ollama, reducing latency by **80%** (2.0 s → 400 ms) — the primary performance metric for the system.
- Developed secure Flask API endpoints serving local LLM responses with 100% on-device execution; integrated OAuth 2.0 and request logging for compliance with sensitive client data handling requirements.

PROJECTS

think-in-silence 2024 – 2025

PyTorch · Transformers · HuggingFace · Python

- Designed a latent reasoning architecture performing K recurrent cross-attention steps entirely in embedding space — no chain-of-thought tokens, no RL training, no reasoning traces at inference time.
- Achieves BLEU 0.231 / ROUGE-1 0.594 at K=8 (2.1× over K=1 baseline) on multi-hop QA; validates K-scaling hypothesis with novel evidence of learned iterative computation.
- Engineered 553K-row synthetic QA dataset via Gemini 2.5 Flash; 3-stage JEPA-style MSE + cosine EMA training; published 1.55 GB model to HuggingFace for full reproducibility.

Nexus Workplace AI 2024

React · Node.js · Socket.IO · GCP · Docker · Firebase · Pinecone

- Full-stack real-time collaboration platform with an agentic AI assistant maintaining workspace-level semantic memory across sessions via a RAG pipeline (Pinecone / Mongo Vector).
- Deployed on GCP with GPU-enabled Compute Engine VMs; IAM-secured multi-tenant access; containerised via Docker with CI/CD pipeline for zero-downtime releases.

Privacy-First Autonomous Email Agent²⁰²⁴

Python · Gmail API · Ollama · Gemma 2 (9B) · OAuth 2.0 · Docker

- Fully local email automation using Gemma 2 (9B) via Ollama + RAG; cross-references Google Calendar API before drafting responses — zero cloud data transmission.
- OAuth 2.0 CLI Human-in-the-Loop verification before every send; containerised via Docker for cross-environment reproducibility.

Visual Deep Research²⁰²⁵

Python · LangGraph · LLaVA · Neo4j · ChromaDB · OCR

- Multimodal autonomous research engine extending text-only deep research to process charts, diagrams, and paper figures using Vision Transformers and LLaVA via Ollama.
- Cross-modal RAG: ChromaDB dense retrieval + Neo4j knowledge graph, orchestrated via LangGraph for multi-step agentic workflows with OCR preprocessing and iterative knowledge synthesis.

CERTIFICATIONS

- **TensorFlow 2.0: Deep Learning** — Udemy | Neural Networks, CNNs, RNNs, Transfer Learning
- **Microsoft AI Fundamentals (AI-900)** — Microsoft | ML Concepts, Azure Cognitive Services, Responsible AI
- **Generative AI Professional Series** — LinkedIn Learning | Prompt Engineering, LLM Workflows, GenAI Applications